

Miguel Contreras

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science, Data Science

Minor in Education

Berkeley, CA

Expected May 2028

Coursework: Machine Learning; Data Science; Computer Architecture; Data Structures and Algorithms; Probability

EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Amazon

Redmond, WA

- Shipped the first health/usage dashboard for the Amazon Leo Enterprise SDK, collapsing 20+ manual CloudWatch/Athena queries across 3 accounts and regions into a sub-2s load over 100+ APIs.
- Built real-time views over 3M+ gateway request log rows into CloudWatch, producing real-time views of API call volume, error attribution, and SDK version adoption by customer account and operation.
- Shipped an agent that triages a CloudWatch alarm on the SDK's request path, classifying the failure to its layer (client, gateway, backing service), tracing it to the owning service and code, and checking for an existing ticket before drafting one.

Software Engineer Contract

Jan 2026 – May 2026

Google

Mountain View, CA

- Evaluated Google's GenAI tools (AI Studio, AlloyDB, Cloud Run) by building a full-stack app to surface developer friction, filing issues to Google's source repos and presenting findings to engineering teams.

Software Engineer Intern

May 2025 – Aug 2025

Amazon

Redmond, WA

- Architected and delivered a full-stack data retention query management system that transformed manual 48-72 hour investigative workflows into automated processes completing in 5-10 minutes, serving global customers.
- Deployed and fine-tuned LLM-powered applications on AWS Bedrock to translate natural language into optimized SQL queries, using retrieval-augmented generation (RAG) for search and data access.
- Integrated 4+ AWS services (Athena, S3, Bedrock, GLUE) into a secure, production-ready architecture with real-time query monitoring, automated file handling, cross-account authentication, and compliance with international data protection regulations.

Data Analyst

Jan 2025 – May 2025

Bay FC — National Women's Soccer League

San Jose, CA

- Built an unsupervised pipeline applying PCA (15 components, 89% variance) and KMeans on 20+ player tracking features to classify players into 6 tactical archetypes at 0.82 silhouette.
- Engineered player-level features (off-ball movement, sprint patterns, line-breaking actions) to predict user engagement-like metrics, aligning with CTR/CVR-style prediction and ranking tasks.

PERSONAL PROJECTS

Refill! – *Typescript, Google AlloyDB, Gemini API, React, PostgreSQL, Docker*

- Designed, built, and deployed a production social app that lets users find, submit, and review water refill stations on an interactive Google Maps interface, with Google OAuth sign-in, photo uploads, a personalized activity feed, mutual friend graph, and refill leaderboard.

Acne Detection & Cross-Domain Classification – *Python, PyTorch, YOLOv5*

- Trained and benchmarked 3 object detection architectures (YOLOv5, Faster R-CNN, DINO-DETR) on the ACNE04 dataset (1,450 images) to localize acne lesions, evaluating models on mAP, precision/recall, and IoU.
- Engineered a cross-domain classification pipeline by extracting positive/negative patches from bounding box annotations and training a binary classifier to generalize from ACNE04 to the DermNet dataset

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, RISC-V, Rust, Scheme

Frameworks: Pytorch, TensorFlow, Tailwind, React, Node.js, FastAPI, OpenCV, Pandas, Numpy

Developer Tools: Git, Google Cloud, Linux, AWS, IAM, Databases, Glue, S3 Buckets, Athena